Lab Sheet 3

Neural Coding: From Spikes to Information

Learning outcomes.

By the end of this lab, you should be able to:

- Explain the difference between rate coding and temporal coding
- Simulate spike trains using probabilistic models
- Understand how refractory periods limit firing rates
- Analyze inter-spike intervals (ISI)
- Demonstrate how population size improves decoding accuracy
- Interpret how biological constraints shape neural information processing

Introduction

In the previous labs, we explored:

- **Lab 01:** Real EEG signal analysis and rhythm extraction
- **Lab 02:** Biophysical origins of EEG using dipole simulations

In this lab, we move one level deeper — from population-level brain signals to the **neuronal code itself**.

Neurons do not communicate using continuous analog signals. Instead, they transmit information using **discrete electrical events called action potentials (spikes)**. These spikes follow the *all-or-none principle* —

- their amplitude (spike size) does not vary
- **timing and frequency do**.

This raises a fundamental neuroscience question:

How do neurons encode information using spikes?

This lab investigates the computational and biophysical principles behind neural coding.

Download the **Lab03_Spike_Train_Modeling_and_Decoding.ipynb** notebook file and modify as follows as commented the notebook file.

TODO 1:

- Change rate = 5, 20, 50 and observe the raster + estimated rate.
- Add a rate estimator using a sliding window (students implement a simple moving average).

TODO 2:

- Vary refrac_s (2ms, 5ms, 10ms) and plot achieved firing rate vs refractory.
- Compute ISI (inter-spike interval) distribution and show the “gap” caused by refractory. (You must write the ISI code.)
- What happens when refactor increases?

TODO 3:

- Modify the encoding model: make it nonlinear (e.g., sigmoid or saturation).
- Add noise to stimulus and check stability of rate estimation.
- Explain what happened to the Raster plot and rate estimation. (1-2 sentences)

TODO 4:

- Implement a coincidence detector: count how often spikes fall within a window of a target time.
- Plot detection rate vs jitter.

TODO 5:

- Vary population size N: test N = 5, 10, 20, 50 and plot accuracy vs N.
- Add shared noise (correlation) and observe decoding accuracy drop.

Writing questions.

1. Rate coding vs timing coding (2–3 sentences)
2. What refractory changes
3. How population size affects decoding

Submission:
Rename your Python notebook with your studentID (**Lab03_Spike_Train_Modeling_and_Decoding.**StudentID .ipynb) and upload.